Online Uncertainty Quantification for the Return-to-Home Decision of a Drone: Strictly Proper Scoring, Polynomial Chaos and Kalman Filtering

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Abstract

A drone triggers its automatic return when the remaining energy no longer covers the predicted return energy plus a margin $m = \kappa E + c$. Read as a Gaussian forecast, this rule has a spread that depends on the flight only through E. We build a flight-conditional predictive law of the arrival state of charge Y. Under explicit assumptions the prediction loop has exactly five exogenous random inputs and $Y = G(\boldsymbol{\xi})$, $\boldsymbol{\xi} \sim \mathcal{N}(0, \mathrm{I}_d)$. Each source is modelled as $\mu + F\boldsymbol{\xi}$ with a germ frozen over the return, which forces the model to expose F rather than FF^{T}. Twenty-one candidate models are ranked by a logarithmic CRPS skill score that splits exactly into a centre and a width term, the latter bounded by $-\log_{10}\operatorname{erf}\sqrt{\ln 2/2} \approx 0.2256$. On board, G is propagated by an unscented transform and by a Hermite polynomial chaos whose coefficients are learnt online by a Kalman filter, with a three-term bound on the decision quantile. On 132 returns the quantile rule saves 0.58 SoC points of margin at equal survival; we prove this smallness is structural.

1. Decision problem

Let $(\Omega, \mathcal{A}, \mathbb{P})$ carry the telemetry filtration $(\mathcal{F}_t)$. The return starts at the stopping time $\tau = \inf\{t : E_{\mathrm{batt}}(t) - E(t) - m(t) \le 0\}$. A *predictive law* is an $\mathcal{F}_\tau$-measurable random probability measure $\hat{F}$ on $\mathbb{R}$; the target is $\mathcal{L}(Y \mid \mathcal{F}_\tau)$. With $q_\alpha(Y) = \inf\{t : F_Y(t) \ge \alpha\}$,

$$q_\alpha(Y) \ge s \;\Rightarrow\; \mathbb{P}(Y \ge s) \ge 1 - \alpha, \tag{1}$$

with equivalence when F_Y is not flat at level α; the converse fails for atoms. The probabilistic rule triggers when $q_\alpha(\hat{F}_t) \le s_{\min}$, i.e. uses the margin $m = \hat{y}_{\mathrm{det}} - q_\alpha(\hat{F}_t)$. The quantile is elicitable: it minimises $x \mapsto \mathbb{E}\rho_\alpha(Y - x)$, $\rho_\alpha(u) = u(\alpha - \mathbb{1}_{\{u<0\}})$ [12].

Germ representation. Tracing the firmware loop (guidance, wind model, speed map, power, current, battery) shows that uncertainty enters only through five sources: gauge error $\varepsilon_{\mathrm{SoC}}$, position error $\boldsymbol{\varepsilon}_p \in \mathbb{R}^2$, wind error $\boldsymbol{W} \in \mathbb{R}^2$, power residual δ and ground-speed error η. The Jacobian of the five loop channels with respect to them is block-diagonal with non-zero blocks, so no source is a function of the others, and omitting one lowers the Gaussian-linearised q_α by $(\sigma - \sigma')|z_\alpha| > 0$, i.e. the forecast becomes optimistic. Hence $Y = G(\boldsymbol{\xi})$, G Borel, $G \in L^2(\gamma_d)$, $d \in \{7, 11\}$.

2. Uncertainty model

Definition 1. *A model is (p, d, μ, F) with $\mu : \mathcal{S} \to \mathbb{R}^p$, $F : \mathcal{S} \to \mathbb{R}^{p\times d}$ on the state space $\mathcal{S}$; $X(s) = \mu(s) + F(s)\boldsymbol{\xi}$, so $X(s) \sim \mathcal{N}(\mu, FF^{\mathsf{T}})$; $d = 0$ is a deterministic reference.*

This is the degree-one Hermite chaos, i.e. the $L^2(\gamma_d)$ projection of any $H(\boldsymbol{\xi})$ on affine maps. The germ is drawn *once* per return: a wind or gauge error persists. Then, along states $s_1, \dots, s_N$,

$$\bar{X} = \bar{\mu} + \bar{F}\boldsymbol{\xi}, \quad \bar{F} = \tfrac{1}{N}\textstyle\sum_k F(s_k), \tag{2}$$

so $\sigma(\bar{X}) = \bar{\sigma}$ in 1-D, whereas independent steps would give $\mathrm{rms}(\sigma)/\sqrt{N}$, up to $\sqrt{N}$ ($10.95\times$ for $N = 120$) too narrow. Two square roots of the same Σ_k may aggregate differently ($F_1 = \sigma\mathrm{I}$, $F_2 = -\sigma\mathrm{I}$ give $\bar{F} = 0$), so the model must expose F, taken as the unique symmetric PSD root of $\Sigma(s)$. An unknown spread is `NaN`, never 0: $\sigma \to 0$ yields the finite score $|y - \mu|$ of a perfectly confident forecast.

3. Strictly proper scoring and the CRPS

For a score $S(\hat{F}, y)$ let $S(\hat{F}, G) = \mathbb{E}_{Y\sim G} S(\hat{F}, Y)$. S is *proper* on a class $\mathcal{P}$ if $S(G, G) \le S(\hat{F}, G)$ for all $\hat{F}, G \in \mathcal{P}$, *strictly proper* if equality forces $\hat{F} = G$ [4]. Hence one criterion both ranks and calibrates.

Definition 2 ([13])**.** *On $\mathcal{P}_1$ (finite first moment),* $\mathrm{CRPS}(\hat{F}, y) = \int_{\mathbb{R}} (\hat{F}(t) - \mathbb{1}_{\{t\ge y\}})^2\, dt$.

Theorem 1. *(i)* $\mathrm{CRPS}(\hat{F}, y) = \mathbb{E}|X - y| - \frac{1}{2}\mathbb{E}|X - X'|$, $X, X' \overset{\mathrm{iid}}{\sim} \hat{F}$. *(ii)* $\mathrm{CRPS}(\hat{F}, y) = 2\int_0^1 \rho_\alpha\big(y - \hat{F}^{-1}(\alpha)\big)d\alpha$. *(iii)* $\mathbb{E}_G\,\mathrm{CRPS}(\hat{F}, Y) - \mathbb{E}_G\,\mathrm{CRPS}(G, Y) = \int(\hat{F} - G)^2$, *so the CRPS is strictly proper on $\mathcal{P}_1$.*

Proof. (iii) Expand the square: $\mathbb{E}(\hat{F}(t) - \mathbb{1}_{\{t\ge Y\}})^2 = (\hat{F} - G)^2 + G(1 - G)$ pointwise, integrate (Fubini). (i) uses $\int(\mathbb{1}_{\{t\ge a\}} - \mathbb{1}_{\{t\ge b\}})^2 dt = |a - b|$; (ii) follows from the change of variables $t = \hat{F}^{-1}(\alpha)$ [5]. □

By (ii) the CRPS averages the pinball losses of all quantiles, the decision functional of (1). Unlike the log score it is finite for point masses and 1-homogeneous; unlike the squared error it sees the spread. For $\sigma > 0$, $z = (y - \mu)/\sigma$ [9],

$$\mathrm{CRPS}(\mathcal{N}(\mu, \sigma^2), y) = \sigma\Big(z \operatorname{erf}\tfrac{z}{\sqrt{2}} + \sqrt{\tfrac{2}{\pi}}e^{-z^2/2} - \tfrac{1}{\sqrt{\pi}}\Big), \tag{3}$$

jointly convex in (μ, σ) with Hessian $\sigma^{-1}\sqrt{2/\pi}e^{-z^2/2}(1, z)^{\mathsf{T}}(1, z)$, minimised in σ at $\sigma^\star = |y - \mu|/\sqrt{\ln 2}$.

Log skill score. With a deterministic reference $\mathrm{CRPS}_{\mathrm{det}} = \sum_i |y_i - \mu_{\mathrm{det},i}|$ and marginal sum CRPS_p, let $\mathrm{crpsl} = -\log_{10}(\mathrm{CRPS}_p \,/\, \mathrm{CRPS}_{\mathrm{det}})$. $-\log$ is the only decreasing scale additive over ratios (Cauchy equation), hence antisymmetric (± 0.301 for a factor 2). With $e = \sum_i |y_i - \mu_i|$,

$$\mathrm{crpsl} = \underbrace{-\log_{10}\tfrac{e}{\mathrm{CRPS}_{\mathrm{det}}}}_{\text{centre}} + \underbrace{-\log_{10}\tfrac{\mathrm{CRPS}_p}{e}}_{\text{width}}. \tag{4}$$

Theorem 2 (width ceiling)**.** *For all μ, y with $e > 0$ and all F,* $\text{width} \le -\log_{10}\operatorname{erf}(\sqrt{\ln 2/2}) \approx 0.2256$, *with equality in 1-D iff $\sigma = \sigma^\star$.*

Proof. At $\partial_\sigma = 0$, $e^{-z^2/2} = 2^{-1/2}$, $z^2 = \ln 2$ and (3) reduces to $|y - \mu|\operatorname{erf}(z/\sqrt{2})$; sum the coordinates. □

The ceiling is the one number comparable across sources. Weights are fitted by $\min_w |\mathcal{T}|^{-1}\sum_{i\in\mathcal{T}}\mathrm{CRPS}(\mathcal{N}(\mu_i,\sigma_i(w)^2),y_i)$ on a persisted 80 % split of flights, convex when σ_i is affine in w; in-sample scores are optimistic, $\mathbb{E}J_n(\hat w)\le\inf J\le\mathbb{E}J(\hat w)$, and two methods are only compared on their common events.

4. Propagation

q_α of $G(\boldsymbol\xi)$ has no closed form even for $G=\xi_1\xi_2$ (density $\pi^{-1}K_0$). Monte Carlo needs $N=\alpha(1-\alpha)/(f(q_\alpha)^2\varepsilon^2)\approx 8\,200$ calls/s for $\varepsilon=0.1$ point [17]; the on-board budget is ≈ 8.

Unscented transform [10, 18]. With $\lambda=0$ ($\alpha_{\mathrm{UT}}=1,\kappa=0,\beta=2$): $\Xi_{\pm i}=\pm\sqrt{d}\,\boldsymbol e_i$, $W_i^m=W_i^c=1/(2d)$, $W_0^c=2$. The rule integrates every polynomial of degree ≤ 3 exactly against γ_d and returns mean and covariance exactly for affine G, at $2d+1$ calls; the predictive is $\mathcal{N}(\mu_{\mathrm{UT}},\sigma^2_{\mathrm{UT}})$.

Polynomial chaos. With normalised Hermite $h_{n+1}=(xh_n-\sqrt{n}\,h_{n-1})/\sqrt{n+1}$ and $\Psi_{\boldsymbol a}=\prod_j h_{a_j}(z_j)$, $\{\Psi_{\boldsymbol a}\}$ is a Hilbert basis of $L^2(\gamma_d)$ [2, 3, 19]: $G=\sum_{\boldsymbol a}c_{\boldsymbol a}\Psi_{\boldsymbol a}$, $\mathbb{E}G=c_{\boldsymbol 0}$, $\operatorname{Var}G=\sum_{\boldsymbol a\neq\boldsymbol 0}c_{\boldsymbol a}^2$, and the truncation residual $R_p=\sum_{|\boldsymbol a|_1>p}c_{\boldsymbol a}^2\to 0$. Sobol indices follow from the Hoeffding decomposition [6, 16]: $S_k=V^{-1}\sum_{\operatorname{supp}\boldsymbol a\subseteq\mathcal{G}_k}c_{\boldsymbol a}^2$. The battery cut-off makes G a step: for $G=\beta\mathbb{1}_{\{z\le 0\}}$, $R_p\sim\beta^2/(\pi\sqrt{2\pi p})$ and R_2 is $1-2/\pi=36\%$ of the variance. Clipping $X^+=\max(X,0)$ lowers the CRPS for $y\ge 0$ and gives $q_\alpha(X^+)=\max(q_\alpha(X),0)$, so the quantile stays exact at the floor.

Theorem 3 (three-term quantile error)**.** *If G, G_p, $\hat G_p$ have density $\ge f_{\min}$ near q_α and $\mathcal{T}(\Delta^2)=\frac{3}{2}(2\Delta^2/f_{\min})^{1/3}$,*

$$|\hat q_\alpha^N-q_\alpha(G)|\le\mathcal{T}(R_p)+\mathcal{T}(\|\hat{\boldsymbol c}-\boldsymbol c\|^2)+D_N/f_{\min},$$

D_N the Kolmogorov discrepancy of the N quadrature nodes.

Proof. Triangle inequality through $q_\alpha(G_p)$, $q_\alpha(\hat G_p)$; Parseval turns L^2 errors into R_p and $\|\hat{\boldsymbol c}-\boldsymbol c\|^2$; a Markov bound on $\{|X-X'|>\eta\}$ optimised in η gives $\mathcal{T}$. □

The nodes are a Sobol sequence [15, 8] mapped by $z=\sqrt{2}\,\mathrm{erf}^{-1}(2u-1)$; Koksma–Hlawka [14] fails for an indicator, but the measured equivalent sample size is 2.4×.

5. Online learning: Kalman filter on coefficients

Truncation gives the *linear* observation $G(\boldsymbol z)=\Psi(\boldsymbol z)^{\mathsf T}\boldsymbol c+\varepsilon_{\mathrm{tr}}(\boldsymbol z)$, $\operatorname{Var}\varepsilon_{\mathrm{tr}}=R_p$, $\varepsilon_{\mathrm{tr}}\perp\Psi_{\boldsymbol a}$. We posit the state-space model

$$\boldsymbol c_{i+1}=\boldsymbol c_i+\boldsymbol w_i,\quad y_{i,r}=\Psi(\boldsymbol z_{i,r})^{\mathsf T}\boldsymbol c_i+v_{i,r},\qquad(5)$$

$\boldsymbol w_i\sim\mathcal{N}(0,Q\Delta_i\mathrm{I})$, $v\sim\mathcal{N}(0,R)$. The Kalman update [11] $\nu=y-\boldsymbol h^{\mathsf T}\hat{\boldsymbol c}$, $S=\boldsymbol h^{\mathsf T}\mathrm{P}\boldsymbol h+R$, $\hat{\boldsymbol c}^+=\hat{\boldsymbol c}+\mathrm{P}\boldsymbol h\nu/S$, $\mathrm{P}^+=(\mathrm{P}^{-1}+\boldsymbol h\boldsymbol h^{\mathsf T}/R)^{-1}$ is the exact Gaussian posterior. Two facts make it fit the problem. (a) $\mathbb{E}[\|\hat G_p-G_p\|^2_{L^2}\mid\text{data}]=\operatorname{tr}\mathrm{P}$: the posterior covariance *is* the epistemic part of the error, distinct from the aleatoric spread $\sum_{\boldsymbol a\neq\boldsymbol 0}\hat c_{\boldsymbol a}^2$. (b) P^+ does not depend on y, so the optimal adaptive design equals the optimal non-adaptive one; germs are chosen greedily by the exact quantile-variance reduction $(\boldsymbol\gamma^{\mathsf T}\mathrm{P}\boldsymbol h)^2/S$, $\boldsymbol\gamma=\nabla_{\boldsymbol c}q_\alpha=\mathbb{E}[\Psi(\boldsymbol\xi)\mid\Psi^{\mathsf T}\boldsymbol c=q_\alpha]$ [7]. Replacing $\sum_r\boldsymbol h\boldsymbol h^{\mathsf T}$ by $n\mathrm{I}$, $\mathrm{P}=(R/n)u\mathrm{I}$ with $u_{i+1}=((u_i+\varrho)^{-1}+1)^{-1}$, $\varrho=Q\Delta t\,n/R$: a tuning (Q,R,P_0) does not transfer across call rates. Normalised innovations $\nu^2/S\sim\chi^2_1$ under the model diagnose mis-tuning [1]; if the candidate pool spans H with $\dim H<P=\binom{d+p}{p}$, directions in $H^\perp$ keep their prior variance whatever the number of calls.

6. Evaluation and why the gain is small

On a flown return, survival of criterion $s=\hat y_{\mathrm{det}}-m$ is $\Phi((y-s)/\sigma_{\mathrm{obs}})$, σ_{obs} the gauge uncertainty at arrival. Rules are compared at equal mean survival; the gain is the mean margin saved. On $n=132$ completed returns the chaos rule saves +0.58 point ($\alpha^\star\approx 0.10$), +0.98 on long returns, the UT +0.04. Per flight the allocation is right: as $|\partial Y/\partial w|=\bar PD/((v^\star-w)^2J)$, a headwind gets more margin, a tailwind less.

Structural bound. Let the firmware error satisfy $\varepsilon\mid x\sim\mathcal{N}(b(x),s(x)^2)$, $\tau^2=s^2+\sigma^2_{\mathrm{obs}}$. If b,τ are constant, Jensen on the concave branch of Φ (rules with $b+m\ge 0$) shows that a constant margin is optimal for any target mean survival $\ge 1/2$. For independent b,τ with variances v_b,v_τ, the constant margin matching the quantile rule $m_q=-b+z\tau$ satisfies

$$m_c-\mathbb{E}m_q=\tfrac{z}{2\bar\tau}\big(v_b+v_\tau(z^2-2)\big)+o(v_b+v_\tau).$$

The gain is second order; heteroscedasticity pays only for $z>\sqrt{2}$ ($\alpha<0.079$), and the always-positive bias term is forfeited by re-centring on $\hat y_{\mathrm{det}}$. The value of conditional uncertainty thus appears under a per-flight risk guarantee or with conditional bias correction.

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